ANIRUDDHA DAS

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SUMMARY

Backend and applied-AI engineer pursuing a dual degree in CS (Adamas University) and Data Science (IIT Madras). Built the backend of Vardaan, a voice AI agent that won 1st place at Aki Hacks × Hacktoberfest ’26, and an ML service that predicts flight arrivals within ~9 min. 6× hackathon winner and finalist (4× first place). Seeking backend or AI/ML internships.

EDUCATION

Indian Institute of Technology Madras *Jul 2024 – Jul 2028*

Bachelor of Science (BS), Data Science & Applications *CGPA: 7.58/10 (Project CGPA: 9.0/10)*

Adamas University *Aug 2024 – Aug 2028*

B.Tech, Computer Science & Engineering *CGPA: 8.0/10*

Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Statistics

PROJECTS

Vardaan – Voice-First AI Agent | 1st Place, Aki Hacks × Hacktoberfest ’26 *Oct 2026*

Backend & AI Engineer, team of 4 | Next.js, TypeScript, Groq/Llama 3.3, Gemini, Sarvam AI | Live: thevardaan.vercel.app | GitHub

- Built the API layer for real inbound and outbound phone calls in 11 languages: a turn endpoint that runs each utterance through the agent for Sarvam’s voice pipeline, and a webhook that stores call outcomes and transcripts.
- Implemented the tool-calling loop (intent parsing, planning, execution) with web search over a Tavily → Brave → DuckDuckGo fallback chain, persistent memory and scheduled reminder callbacks.
- Put Groq (Llama 3.3 70B) and Gemini 2.5 Flash behind one OpenAI-compatible client; secured endpoints with shared secrets and HMAC-signed sessions with scrypt hashing.

Checkpost – Policy Firewall for AI Agents | Hackathon project *Jun 2026*

Backend & Policy Engine | Next.js, TypeScript, Firebase/Firestore, Supabase | Live: checkpost-ai.vercel.app | GitHub

- Built the proxy API that intercepts every AI-agent tool call and returns allow, block or hold before it reaches production.
- Implemented a deterministic, deny-by-default policy engine (YAML rules, tool allow-lists, exact parameter matching) with no model inference on the enforcement path, keeping checks under 1 ms.
- Added cost governance that tracks spend and retries per run and hard-stops runaway agents, plus a human approval queue and audit trail in Firestore.

AeroGrid – Aviation Telemetry & ETA Prediction | Personal project *Jun 2026 – Oct 2026*

React, TypeScript, Three.js, Node.js/Express, FastAPI, PyTorch | GitHub

- Architected a three-service system (React/Three.js, Express gateway, FastAPI ML service) tracking ~253 live aircraft; a one-minute cache with last-good fallback survives OpenSky rate limits and caps upstream calls at one per minute.
- Built a pipeline turning ~29,900 raw ADS-B records (1,557 aircraft) into 6,157 labeled flights via 30-minute track splitting, vectorized Haversine distance to 4 hubs and time-to-arrival labels.
- Trained a PyTorch MLP to ~9.1 min mean ETA error on held-out data, 55% below a mean baseline (R^2 0.67), with 52% of predictions within 5 min and ~0.03 ms inference.

Placement Portal – Resume-to-Job Matching Platform | IIT Madras *Jan 2026 – Jun 2026*

React, Vite, Flask, SQLAlchemy, scikit-learn, Gemini API | GitHub

- Built a Flask REST API and React SPA with session-cookie auth, replacing a server-rendered Jinja prototype.
- Ranked candidates per job with TF-IDF and cosine similarity over parsed PDF resumes, weighting explicit technical terms.
- Generated explainable Gemini evaluations citing resume evidence, cached to avoid repeat API calls.

ACHIEVEMENTS

- **1st Place, Aki Hacks × Hacktoberfest ’26** (MLH, Machine Learning Kolkata): built Vardaan.
- **1st Place, Coding Premier League**, SIGNIFIYA 2026: first of 40+ teams in competitive programming.
- **1st Place, VibeForge 1.0**, Adamas University.
- **1st Runner-Up, InnovateX 2.0**, IMI Kolkata Techfest: Adamas University’s first top-3 finish there.
- **Finalist, Hackforge**, Srijan 2026, Jadavpur University: Top 25 of 1,000+ submissions.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL

Backend & APIs: Node.js, Express.js, FastAPI, Flask, Next.js, REST API design, Webhooks, Authentication (sessions, HMAC)

AI / ML: PyTorch, scikit-learn, pandas, Feature Engineering, TF-IDF, LLM Tool Calling, Groq (Llama 3.3), Gemini API

Frontend: React, Next.js, Tailwind CSS, Three.js, WebGL

Data & Cloud: Supabase, Firebase/Firestore, SQLite, Google BigQuery, Google Cloud Platform, Vercel, Git, Linux

Certifications: Google Cloud Skill Badge, Use ML APIs on Google Cloud (2026); Programming in Java, NPTEL (2026)